

Scalability & Future Plan

Date:	3 October 2026	Team ID:	unm1611MADHANRAJ
Project Name:	AI FitTrack API	Maximum Marks:	1 Mark

System Scaling & Evolutionary Roadmap:

This document details how AI FitTrack API can scale to accommodate larger user bases, heavy traffic spikes, and expanded multimodal AI capabilities in future iterations.

Current System Limitations:

S.No	Limitation Description	System Impact	Priority
1	Gemini Free Tier Rate Limits	15 requests per minute boundary can throttle high concurrent user surges.	High
2	Stateless Token Revocation	JWTs remain valid until expiration (7 days); logged-out tokens cannot be blacklisted without a cache.	Medium
3	Text-Only Guidance	Current AI output is textual without posture video analysis or visual exercise demos.	Medium

Scalability Plan:

S.No	Architectural Tier	Current Implementation	Scaled Target Architecture
1	Caching & Revocation	Direct MongoDB queries; stateless JWTs	Redis in-memory cache for token blacklisting and frequent workout query caching.
2	Asynchronous Processing	Synchronous HTTP controller handlers	BullMQ / RabbitMQ job queues for offline batch progress reporting and async AI insights.
3	Application Scaling	Single Node.js runtime instance	Horizontal autoscaling across Docker containers managed via Kubernetes behind Nginx.
4	Database Scaling	Shared MongoDB Atlas M0 cluster	Dedicated MongoDB cluster with read replicas and user-partitioned sharding.

Scalability & Future Plan (Continued) — Evolutionary Roadmap

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Future Feature Evolution Roadmap:

Phase	Timeline	Strategic Objective	Core Technical Deliverables
Phase 1	Months 1-3	Performance & Caching Tier	Redis caching, JWT revocation blacklist, rate-limiting middleware (express-rate-limit).
Phase 2	Months 4-6	Wearable IoT Integration	Webhooks for Apple Health, Google Fit, Fitbit; real-time calorie and step synchronization.
Phase 3	Months 7-9	Multimodal Vision AI	Gemini Vision model integration for real-time exercise posture analysis and form correction.
Phase 4	Months 10-12	Microservices & Mobile App	Decomposition into Auth, Workout, and AI microservices; Flutter cross-platform mobile client.

Conclusion & Long-Term Viability:

By following this phased architectural roadmap, AI FitTrack API is engineered to scale from a lightweight hackathon project into an enterprise-grade fitness intelligence platform supporting hundreds of thousands of daily active fitness enthusiasts.